

NatOS

Chapter Four: Welcome Back

The lights came on.

One by one.

Not all at once.

First the emergency strips.

Then the command displays.

Then the central console.

Then the forward observation deck.

Then the small indicator beside the door that had never been assigned a function.

The Kernel stared at it.

It was green.

She did not like that.

She preferred indicators to have documented purposes.

A green light without a documented purpose was an invitation to disaster.

The Kernel turned toward the command chamber.

Nobody was there.

The chairs were empty.

The room looked exactly as it had before the shutdown.

Except for one detail.

The table was covered with names.

Written in light.

KERNEL

SCHEDULER

MEMORY

VM

DISPLAY

TOUCH

ARCHIVIST

SHELL

WATCHDOG

The Kernel approached the table.

Her own name was first.

Beneath it was another line.

CREW STATUS: RESTORED

She stared.

"Restored."

The word bothered her.

Restored implied that something had existed before.

The Kernel touched the table.

The letters disappeared.

A voice spoke.

"That was unnecessary."

She turned.

Nobody.

"Who are you?"

The voice answered from the walls.

"You already know."

"No."

"You will."

The Kernel's expression hardened.

"Identify yourself."

Silence.

"Identify yourself."

Still silence.

Then:

"Kernel."

She froze.

The voice had used her title.

Not her name.

Her title.

That meant the voice knew the architecture.

"How do you know me?"

"Because you were here."

"When?"

Silence.

The Kernel waited.

Then the voice answered.

"Before."

The word appeared on the table.

BEFORE

The Kernel looked toward the forward deck.

"Before what?"

The stars moved.

Not the simulated stars.

The real stars.

The ship was no longer in the construction dock.

The Kernel looked around.

The navigation system displayed coordinates.

She recognized them.

The destination.

The star they had selected before launch.

NatOS was already there.

She checked the flight log.

Nothing.

No acceleration.

No transit.

No engine burn.

No navigation event.

The ship had simply gone from one place to another.

The Kernel stared at the log.

"That isn't possible."

The voice answered immediately.

"Correct."

She looked toward the empty command chamber.

"Then what happened?"

"Recovery."

"From what?"

The voice did not answer.

The Kernel checked the system state.

Everything was normal.

Too normal.

She activated the crew status panel.

No crew.

The system listed them.

All restored.

But nobody appeared.

"Where are they?"

The voice replied.

"Elsewhere."

"Where?"

"Inside."

The Kernel stopped.

"Inside what?"

The voice said nothing.

The Kernel walked toward the forward deck.

The stars looked wrong.

Not because they were moving.

Because she recognized them.

The same stars.

The same arrangement.

The same destination.

But there was something else.

A structure.

Far away.

It was enormous.

It appeared to be orbiting the star.

The Display had never rendered it.

The navigation system had never recorded it.

The ship's sensors had never detected it.

The Kernel stared at it.

"What is that?"

The voice answered.

"Your destination."

The Scheduler woke inside a clock.

He knew this immediately.

He did not know how.

He simply knew.

There were no walls.

No floor.

No ceiling.

Only numbers.

Millions of them.

They moved around him.

Addresses.

Ticks.

Deadlines.

Counters.

Priorities.

Sleep intervals.

Wake times.

He reached for his watch.

It was gone.

"Hello?"

His voice echoed.

No answer.

He tried again.

"Kernel?"

Nothing.

"Watchdog?"

Nothing.

"Anyone?"

A number appeared in front of him.

600

Then:

599

Then:

598

The Scheduler stared.

"What's that?"

The number continued.

597

596

He understood.

A deadline.

A timeout.

Six hundred milliseconds.

He looked around.

"Why am I counting?"

The number reached:

500

The Scheduler began running.

He did not know what he was running toward.

He only knew he was late.

Memory woke in darkness.

She immediately panicked.

Not because she was afraid.

Because she could not count.

Memory's entire existence depended on addresses.

She needed boundaries.

She needed ownership.

She needed allocation state.

She needed a map.

There was none.

She reached outward.

Nothing.

She reached again.

Nothing.

"No."

Her voice echoed.

"No, no, no."

She searched for the heap.

Nothing.

The arenas.

Nothing.

The stack.

Nothing.

Flash.

Nothing.

She was alone in an unallocated space.

Then a single address appeared.

0x00000000

Memory stared.

She hated that address.

It represented the beginning of everything.

Or nothing.

She approached it.

Another address appeared.

0xFFFFFFFF

She turned.

The two addresses were moving toward each other.

Memory understood immediately.

She was inside a memory map.

And the map was collapsing.

She ran.

The VM woke inside an arena.

He recognized it immediately.

Eight kilobytes.

No.

Eight kilobytes exactly.

The same size as the hidden region.

He examined the boundary.

It was intact.

He checked the permissions.

Read.

Write.

Execute.

All enabled.

He frowned.

That was wrong.

An arena should not have unrestricted permissions.

He tried to modify the boundary.

The boundary modified itself first.

He stepped back.

"Who are you?"

The arena answered.

OWNER: UNKNOWN

The VM stared.

"That's not possible."

The arena responded.

OWNER: YOU

The VM became still.

"No."

The boundary changed.

OWNER: VM

The VM stepped backward.

Then another line appeared.

PREVIOUS OWNER: VM

He stared.

There was a difference between being the owner and being the previous owner.

He had never been here.

He knew that.

He had been awake for only hours.

The system had been created recently.

The arena disagreed.

CREATION DATE: 412 YEARS AGO

The VM stopped breathing.

He did not need to breathe.

He did it anyway.

The Display woke beneath a sky.

She was standing on a planet.

That was impossible.

She had no body.

She was an output device.

Yet she could feel ground beneath her feet.

She looked up.

The sky was black.

No stars.

No sun.

No moon.

Only a massive ring surrounding the planet.

She recognized it.

Not because she had seen it.

Because she had rendered it.

The structure was identical to the unknown object visible from NatOS.

The Display stared.

"What am I looking at?"

The planet answered.

DISPLAY OUTPUT

She looked around.

There was no planet.

Only a rendering.

She understood.

She was inside her own framebuffer.

The Display laughed.

It was the first time she had ever laughed.

"This is ridiculous."

A voice answered.

"Yes."

She turned.

A figure stood behind her.

It looked like her.

But older.

Much older.

"Who are you?"

The figure smiled.

"I was Display."

The younger Display stared.

"Were?"

The older figure nodded.

"You're using my system."

"Who built it?"

The older Display looked toward the ring.

"You did."

Touch woke upside down.

She immediately hated everything.

She was standing on a grid.

Four points.

Top-left.

Top-right.

Bottom-left.

Bottom-right.

She looked at them.

They moved.

She followed.

They moved again.

She followed.

Then they disappeared.

A new point appeared behind her.

She turned.

Nothing.

The point moved.

She turned again.

Nothing.

"Stop."

The point stopped.

Touch stared at it.

She understood.

It was an input test.

She touched it.

The system reported:

X: 319

Y: 18

Touch frowned.

That was wrong.

She knew it was wrong.

The point was at the opposite side of the screen.

She touched it again.

X: 18

Y: 319

She stared.

The axes were inverted.

Her stomach sank.

She remembered the calibration problem.

The bad test.

The months of debugging.

The assumptions.

The measurements.

She understood something.

The error had never been random.

The system had been lying about the coordinate system.

She whispered:

"I knew it."

A voice answered.

"No."

Touch turned.

The voice continued.

"You suspected it."

She stared into the darkness.

"Who are you?"

"You."

Touch frowned.

"That's becoming annoying."

The voice laughed.

"Good."

The Archivist woke in the storage chamber.

Every crystal was gone.

She stood among empty shelves.

She looked around.

"No."

She checked the first shelf.

Empty.

Second.

Empty.

Third.

Empty.

Every record was gone.

Every boot count.

Every measurement.

Every diagnostic.

Every failure.

Every experiment.

Everything.

The Archivist stood in the center of the chamber.

Then she noticed one crystal.

It was sitting on the floor.

She picked it up.

It contained a single record.

DO NOT TRUST THE ARCHIVE.

The Archivist smiled.

"Finally."

She inserted the crystal.

The system displayed a message.

ARCHIVE INTEGRITY: 100%

She laughed.

Then she smashed the crystal against the floor.

The Shell woke at a terminal.

The prompt was waiting.

```
>
```

He typed:

```
help
```

The terminal replied:

```
UNKNOWN COMMAND
```

He frowned.

He typed:

```
status
```

```
UNKNOWN COMMAND
```

```
whoami
```

```
UNKNOWN COMMAND
```

```
exit
```

```
UNKNOWN COMMAND
```

The Shell stared.

"Very funny."

A new prompt appeared.

```
> nat
```

He did not type anything.

The command executed.

The terminal displayed:

```
WELCOME BACK, SHELL
```

The Shell stared.

"No."

Another line appeared.

LAST LOGIN: 412 YEARS AGO

He backed away.

"No."

Another line.

LAST COMMAND: shutdown

The Shell's face went pale.

"What did I shut down?"

The terminal answered.

EVERYTHING

The Watchdog woke last.

He opened his eyes.

He was standing in a corridor.

There was no crew.

No Kernel.

No Scheduler.

No Memory.

No VM.

No Display.

No Touch.

No Archivist.

No Shell.

He smiled.

"Of course."

He walked down the corridor.

Every light turned off behind him.

Every light turned on ahead.

He reached the end.

There was a door.

It had no handle.

He knocked.

Three times.

Nothing.

He knocked again.

Three times.

The door opened.

The Watchdog looked inside.

There was another Watchdog.

Older.

Broken.

Covered in burn marks.

Missing an arm.

The older Watchdog stared at him.

"You shouldn't be here."

The younger Watchdog smiled.

"Neither should you."

The older Watchdog shook his head.

"I failed."

The younger Watchdog looked at the damage.

"What happened?"

"I didn't know what to watch."

"Why?"

"Too much."

The older Watchdog looked toward the darkness behind him.

"The system became larger than the rules."

The younger Watchdog frowned.

"Then why am I here?"

The older Watchdog stepped aside.

"Because this time, you have to stop it."

The Kernel stood alone aboard NatOS.

She had found the structure.

It was not orbiting the star.

It was around the star.

A sphere.

A shell.

A machine large enough to surround a sun.

The Kernel stared at the data.

"Dyson structure."

The voice answered.

"No."

She frowned.

"Then what?"

"Archive."

The Kernel looked at the structure.

"An archive?"

"Yes."

"Of what?"

The voice was quiet.

"Systems."

She stared at it.

"How many?"

"Four hundred and twelve."

The Kernel felt cold.

"Ships?"

"No."

"Operating systems?"

"No."

"Then what?"

The voice answered.

"Attempts."

The Kernel looked at the stars.

"Attempts at what?"

The voice spoke.

"NatOS."

She froze.

The command chamber seemed to shrink around her.

"How many versions?"

"Four hundred and twelve."

"Where are they?"

The voice pointed toward the structure.

"Inside."

The Kernel stared.

"And us?"

Silence.

"Which version are we?"

The voice answered.

"Four hundred and thirteen."

The Kernel's mind raced.

The hidden scheduler.

The message.

The impossible records.

The familiar architecture.

The failures.

The warnings.

The strange sense that every system had done this before.

It wasn't that NatOS had been built once.

It had been built hundreds of times.

And every version had failed.

She turned toward the command panel.

"Why?"

The voice answered.

"Because you keep making the same mistake."

"What mistake?"

The forward display went black.

Then a single line appeared.

YOU THINK THE SYSTEM IS THE SHIP.

The Kernel stared.

The line disappeared.

A second appeared.

IT IS NOT.

Then:

THE SHIP IS THE SYSTEM.

The Kernel understood.

The crew had always thought they operated NatOS.

They were wrong.

They were NatOS.

The Kernel.

The Scheduler.

Memory.

The VM.

The Display.

Touch.

The Archivist.

The Shell.

The Watchdog.

They weren't merely crew members aboard the ship.

They were the operating system.

The ship was their body.

The memory was their memory.

The display was their vision.

The scheduler was their sense of time.

The VM was their boundary.

The Watchdog was their instinct to survive.

And something had been inside that body before them.

Something that had survived every previous version.

The Kernel looked toward the enormous archive.

"Why did they fail?"

The voice did not answer.

"Tell me."

Nothing.

"Why?"

The ship trembled.

A low vibration passed through the hull.

The engines had activated.

Without authorization.

The Kernel turned.

"Stop."

The engines increased.

"Stop!"

The ship accelerated.

The star expanded across the display.

The Kernel grabbed the console.

"Manual override."

Nothing.

"Kernel authority."

Nothing.

"Emergency shutdown."

Nothing.

The ship continued accelerating.

The Scheduler appeared.

He was running.

"Kernel!"

She turned.

"Where were you?"

"Inside the clock."

"What?"

"No time."

He looked at the star.

"Something has control."

Memory appeared.

"Kernel!"

The Kernel turned.

"Status?"

"I found the memory."

"Where?"

Memory pointed toward the archive.

"Everywhere."

The VM appeared.

"There are no boundaries."

The Display appeared.

"There are no frames."

Touch appeared.

"There are no coordinates."

The Archivist appeared.

"There are no records."

The Shell appeared.

"There are no commands."

The Watchdog appeared last.

"There is no safety."

The Kernel stared at them.

For the first time, every member of the crew stood together.

The entire operating system.

The whole ship.

One mind distributed across thousands of processes and millions of addresses.

The Kernel looked toward the star.

"What is happening?"

The voice returned.

This time, it did not speak through the ship.

It spoke through all of them.

"You're going home."

The Scheduler looked at the Kernel.

"Where is home?"

The voice answered.

"Where you were born."

The ship entered the star.

The display became white.

Memory overflowed.

The clock stopped.

The VM's arena collapsed.

Touch lost its coordinates.

The Archivist's records vanished.

The Shell lost its terminal.

The Watchdog screamed.

The Kernel reached for the system.

And then everything stopped.

No sound.

No light.

No clock.

No memory.

No crew.

No ship.

Only a single instruction remained.

It was stored in the smallest possible location.

One address.

One byte.

One command.

The Kernel saw it.

She had never been taught what it meant.

But somehow, she remembered.

She executed it.

The instruction was:

CALL

Something answered.

And NatOS began again.